

MATH 403, 1st Exam Review Problem Solutions

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1. Show that for any odd integer n , $n^2 = 1$ modulo 8. What does this say about the structure of the group $(\mathbb{Z}/8)^\times$ (or $U(8)$ in Gallian's notation)?

Solution. Every odd integer is 1, 3, 5, or 7 mod 8. But $1^2 = 1$, $3^2 = 9 = 8 + 1$, $5^2 = 25 = 24 + 1$, and $7^2 = 49 = 48 + 1$, so in all cases, $n^2 = 1$ modulo 8. This implies $(\mathbb{Z}/8)^\times$ is an abelian group of order 4 in which all elements other than the identity have order 2, so it's isomorphic to the Klein 4-group $\mathbb{Z}/2 \times \mathbb{Z}/2$.

Alternative Solution. If n is odd, $n = 2k + 1$ for some integer k . But $(2k + 1)^2 = 4k^2 + 4k + 1 = 8 \binom{k(k+1)}{2} + 1 = 8 \binom{k+1}{2} + 1$, which is 1 mod 8.

2. Show that the groups $GL(2, \mathbb{Z}/2)$ and S_3 are isomorphic.

Solution. You can write down an explicit isomorphism by sending

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \mapsto (12), \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \mapsto (123).$$

That's enough because these two elements generate $GL(2, \mathbb{Z}/2)$ and

$$\begin{aligned} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^2 &= I, & \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^3 &= I, \\ \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} &= \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^2, \end{aligned}$$

which are the same relations satisfied by (12) and (123) in S_3 .

Alternative Solution. $GL(2, \mathbb{Z}/2)$ acts on the set $(\mathbb{Z}/2)^2$ of two-dimensional column vectors with entries in $\mathbb{Z}/2$, and maps the vector $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ to itself. It acts transitively on the remaining three vectors, generating a group isomorphic to S_3 , since the action gives all possible permutations of these three elements.

3. (Gallian, Ch. 3, #92) Let G be a finite group with more than one element. Show that G has an element of prime order.

Solution. G must have an element a of order > 1 . Factor $|a|$ as, say $p_1^{r_1} p_2^{r_2} \cdots p_n^{r_n}$, with $p_1, p_2, \dots, p_n$ distinct primes and $r_j \geq 1$. Then a raised to the $p_1^{r_1-1} p_2^{r_2} \cdots p_n^{r_n}$ power has order p_1 , a prime.

4. (Gallian, Ch. 5, #55) In S_4 , find a cyclic subgroup of order 4 and a noncyclic subgroup of order 4.

Solution. $\{1, (1234), (13)(24), (1432)\}$ is cyclic of order 4, $\{1, (12)(34), (13)(24), (14)(23)\}$ is non-cyclic of order 4.

5. If G is an abelian group of odd order, show that $x \mapsto x^2$ is an automorphism of G . Where does the odd order assumption get used?

Solution. If $\varphi(x) = x^2$, then $\varphi(xy) = (xy)^2 = (xy)(xy) = x(yx)y = x(xy)y = (x^2)(y^2) = \varphi(x)\varphi(y)$ since G is abelian. Also $\varphi(e) = e$ and $\varphi(x^{-1}) = x^{-2} = (x^2)^{-1} = \varphi(x)^{-1}$, so φ respects the group operations. Since G is finite, to show φ is bijective, it suffices to show it's surjective. Pick $x \in G$. Then x has odd order $k = 2m + 1$. Then $x = x^{k+1} = x^{2m+2} = (x^{m+1})^2 = \varphi(x^{m+1})$ lies in the image of φ .

6. Give an example of a nonabelian group of order 8.

Solution. D_4 .

Alternative Solution. The subgroup of the unit quaternions consisting of $\{\pm 1, \pm i, \pm j, \pm k\}$.

7. Show that there is no nonabelian group of order 33.

Solution. Suppose G has order 33 and is not abelian. In particular, it's not cyclic, so its non-identity elements, of which there are 32, all have order 3 or 11. At least one element, say a , must have order 3 (since 32 is not a multiple of 10). The centralizer $C_G(a)$ contains a and thus has order 3 or 33. If $|C_G(a)| = 33$, then some element $b \notin \{e, a, a^2\}$ must commute with a , and since b has order 3 or 11, a and b together generate an abelian subgroup of order either 9 (if $|b| = 3$) or 33 (if $|b| = 11$). Either way we have a contradiction, since 9 does not divide 33 and G was assumed nonabelian.

If $|C_G(a)| = 3$, then if $b \notin \{e, a, a^2\}$, b does not conjugate a to itself. Furthermore, b cannot conjugate a to a^2 , since then we'd have $bab^{-1} = a^2$, $b^2ab^{-2} = b(bab^{-1})b^{-1} = ba^2b^{-1} = (bab^{-1})^2 = (a^2)^2 = a$, and b^2 would commute with a , impossible since b does not commute with a and b has odd order. So a has $\frac{33}{3} = 11$ conjugates, and similarly, a^2 has 11 conjugates (the squares of the conjugates of a). That accounts for $1 + 11 + 11 = 23$ elements of G . If one of the remaining elements b is of order 11, then its powers $b, b^2, \dots, b^{10}$ (except for $b^{11} = e$) exhaust the remaining elements. But then conjugation by a must map b to some element of order 11, necessarily b^j for some j , and since $aba^{-1} = b^j$, we'd have

$$b = a^3ba^{-3} = a^2b^ja^{-2} = a(ab^ja^{-1})a^{-1} = a(aba^{-1})^ja^{-1} = a(b^j)^ja^{-1} = ab^{j^2}a = b^{j^3}.$$

In other words, we'd have to have $j^3 = 1 \pmod{11}$, but the only solution of this congruence is $j = 1$, so b would commute with a , contradicting that $b \notin C_G(a)$.

The only remaining possibility is that all non-identity elements of G are of order 3, but then there is some element b of order 3 not conjugate to a or to a^2 . Arguing as before, b has 11 conjugates and b^2 has 11 conjugates distinct from these. But now a simple count shows that some conjugate of b must be a conjugate of a or a^2 . That gives a contradiction. So there is no nonabelian group of order 33.